

DANG TRONG PHUC

Game Backend Engineer · C++ / Java · Real-Time Systems · High-Concurrency Server
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PROFESSIONAL SUMMARY

Game Backend Engineer with 2+ years of professional experience building high-concurrency, low-latency game server systems in C++ and Java. Specializes in authoritative server architecture, real-time state synchronization, matchmaking, session lifecycle management, and concurrent room design for mobile multiplayer games (card battle, MOBA/MMO). Proficient in gRPC/Protobuf communication, Redis-centric data layers (session, leaderboard, pub/sub), and event-driven microservices (Kafka, RabbitMQ). Hands-on with thread-safe concurrency (mutex, thread pools, atomic ops, lock contention minimization), performance profiling, and stress testing at 10,000+ concurrent users. Proven in Agile cross-functional teams; collaborative with client, QA, and DevOps engineers.

TECHNICAL SKILLS

Languages: C++ (C++11/14/17, STL, multithreading, memory management); Java 8/21; SQL

Game Server: Authoritative Server Design, Real-Time State Sync, Round Lifecycle Control, Matchmaking Engine, Room/Session/Player State Management, Thread-Safe Concurrency, Race Condition Prevention, Exploit Prevention

Concurrency (C++): Mutex, Thread Pools, Atomic Operations, Lock Contention Minimization, Memory-Efficient Data Structures, CPU Profiling

Real-Time Comms: gRPC / Protocol Buffers (schema design, versioning), WebSocket, Socket Programming

Backend (Java): Spring Boot 3, Spring MVC, Spring Security + JWT, Spring Data JPA/Hibernate, RESTful API

Data & Caching: Redis (session, leaderboard via Sorted Sets, pub/sub, rate-limiting, distributed locking), MySQL, PostgreSQL

Messaging: Apache Kafka, RabbitMQ (async game event pipelines, background task processing)

Observability & Perf: Structured logging, latency benchmarking, load testing (10k+ virtual users), root cause analysis, memory/CPU profiling

DevOps & Tools: Docker, Linux (Ubuntu/CentOS), Git, GitLab, Maven, Postman, Jira

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

Sep 2018 – Dec 2023

Bachelor of Engineering · Computer Electronic Engineering · Ho Chi Minh City

WORK EXPERIENCE

GIHOT STUDIO · Game Backend Engineer (C++ & Java)

04/2024 – Present · Ho Chi Minh City

Building production backend systems for real-time multiplayer mobile games in a 20+ engineer Agile team. Responsible for game server logic, real-time synchronization, internal tooling, and performance optimization.

- Designed and implemented **server-authoritative game logic** in C++: round lifecycle, matchmaking engine, result calculation, reward distribution, and mission tracking systems for 5,000+ concurrent sessions.
- Engineered **thread-safe room/player state management** using mutex locks, thread pools, and atomic operations; eliminated race conditions under high-concurrency loads, ensuring consistent and fair gameplay outcomes.
- Built gRPC services with Protobuf schemas (auth, matchmaking, in-game events, state sync) achieving sub-50ms client-server RTT; diagnosed and resolved complex sync bugs via systematic root cause analysis.
- Optimized Redis caching (cache-aside, Sorted Sets for leaderboards, pub/sub for real-time events) and RabbitMQ async pipelines — reduced DB load by ~30% and API response time by ~20%.
- Stress-tested system at 10,000+ virtual concurrent users; performed memory/CPU profiling, identified bottlenecks, applied concurrency tuning — achieved 25% end-to-end latency reduction.
- Built Spring Boot (Java) microservices for GM Tool admin dashboard, game config management, and real-time monitoring; collaborated across QA, DevOps, and client teams in GitLab/Jira Agile sprints.

Stack: C++17, Java 8/21, Spring Boot, gRPC, Protocol Buffers, Redis (pub/sub, sorted sets, distributed lock), RabbitMQ, MySQL, Docker, Linux, GitLab, Jira

HAHALOLO TRAVEL SOCIAL NETWORK · Java Backend Developer

06/2022 – 05/2023 · Ho Chi Minh City

Part-time backend developer during final university years. Built RESTful APIs, security systems, and database optimization for a high-traffic social travel platform.

- Designed RESTful APIs using Java 8, Spring Boot, Spring MVC, and Spring Data JPA; implemented Spring Security + JWT with RBAC for endpoint security and access control.

- Profiled and optimized SQL queries and JPA/Hibernate data access layers — reduced average query latency by ~40%; ensured data consistency across services.
- Collaborated with QA and DevOps in Docker/Linux environments; participated in production incident debugging and rapid resolution.

Stack: Java 8, Spring Boot, Spring Security, JWT, JPA/Hibernate, MySQL, Docker, Linux, Git, Maven

Self-directed period (06/2023 – 04/2024): Completed final-year thesis and graduated PTIT (Dec 2023); used remaining time for personal backend projects and self-directed study in C++ game server and distributed systems before joining Gihot Studio.

KEY PROJECTS

Dau Truong Huyen Thoai — Confidential MOBA/MMO Mobile Game (C++ Backend)

02/2025 – Present

Role: Game Backend Engineer · Company: Gihot Studio

- Architected **high-performance MOBA/MMO backend** using C++17 and event-driven architecture; designed full Protobuf schema suite covering authentication, matchmaking, in-game economy (virtual currency/item transactions), hero/player stats, and real-time state synchronization.
- Built Redis-centric data layer: real-time session management, leaderboard systems (Sorted Sets), rate-limiting, pub/sub messaging for horizontal scaling without DB bottlenecks.
- Developed **thread-safe server logic** with focus on CPU efficiency, lock contention minimization, and memory-efficient data structures; enforced strict data consistency for virtual currency transactions under concurrent load.
- Performed latency benchmarking, stress testing (10k+ virtual users), and structured performance profiling; built logging/monitoring observability stack for fast production diagnosis.

Stack: C++17, gRPC, Protocol Buffers, Redis (pub/sub, sorted sets, rate-limiting), Event-Driven Architecture, Linux server environment

Sieu Anh Hung Dai Chien — Real-Time Card Battle Mobile Game (Java Backend)

04/2024 – 11/2024

Role: Game Backend Engineer · Company: Gihot Studio

- Built **microservices backend** for real-time card battle game using Java/Spring Boot; reduced system bottlenecks by 25% and optimized for 5,000+ concurrent users in load tests.
- Engineered high-concurrency socket layer with Protobuf serialization; implemented reward distribution, mission tracking, and session handling engines with strict state consistency.
- Developed GM Tool REST APIs for live game configuration, player data management, and real-time operational monitoring; collaborated with QA on feature testing and post-deployment stability.

Stack: Java 8, Spring Boot, Spring Security, JWT, Protocol Buffers, Redis, MySQL, Socket, Docker, Linux

FPM — Financial Portfolio Manager (Personal Microservices Project)

10/2025 – Present

- Architected 10-service cloud-native system (User-Auth, Wallet, Transaction Engine, Reporting, Notification, OCR, AI, API Gateway, Eureka, Config Server) — demonstrates distributed systems depth beyond game backend.
- Dual-broker event pipeline: Kafka (transaction streaming) + RabbitMQ (domain events); gRPC inter-service layer with 6 Protobuf contracts; Redis multi-layer caching.
- Applied Resilience4j Circuit Breaker, rate limiting, and dual-layer JWT security; Docker Compose + GitLab CI/CD.

Stack: Java 21, Spring Boot 3.5, Spring Cloud, Kafka, RabbitMQ, Redis 7, PostgreSQL, gRPC/Protobuf, Docker Compose, GitLab CI

ADDITIONAL INFORMATION

English: B1 Aptis Certificate (British Council) — actively improving to B2; reads technical documentation and communicates in English-speaking game development teams.

Methodology: Agile / Scrum · GitLab + Jira · sprint planning, daily standup, retrospective, cross-team collaboration.

Engineering focus: Real-time system reliability, concurrent programming, performance profiling, distributed architecture, low-latency server design.